

Lecture 5: Sector coupling and transmission

Energy Economics of the Green Transition

Rasmus K. S. Berg

Monday 5 October 2026

Department of Economics, University of Copenhagen

Today

1. Where we left things: the two tails of the balancing problem
2. Trade over wires
3. Electrification and heat
4. A workshop on Ruhnau et al. (2020): do heat pumps help integrate wind?



Today

Timetable: three blocks of 45 minutes, breaks of 15. The first block is lectured; the workshop has two parts, one per block, each with 30 minutes of work.

block	what
1	Where we left things Trade over wires Electrification and heat; the workshop's question
break	
2	Part 1 setup **Work 1**, 30 minutes **Solutions 1**
break	
3	Part 2 setup **Work 2**, 30 minutes **Solutions 2** Wrap-up

Where we left things

Where we left things

- **By levelised cost**, wind and solar are the cheapest plants we can build (lecture 4).
- **Their value falls as they grow** (lecture 3). The problem is not what they cost but when they produce: every hour, supply must meet demand.
- **Four ways to balance the hour:** flexible backup, storage, flexible demand, trade.
- **Storage** can work on both tails of the problem, at a cost set by the duration: hours (batteries), days (pumped hydro), seasons (reservoir hydro).
- **Today:** trade, and flexible demand from electrified heat.

Where we left things



Lead line: last week ended on the balance: cheap renewables are worth less the more there are, unless some other term of the hourly balance can move.

- **Cannibalisation:** in lecture 4's opening, wind's value fell below its levelised cost at around 40–45% of consumption.
- **Storage is two capacities,** power and energy, paid by different hours; curtailment is its partner, and a skewed surplus keeps the tank small (Zerrahn et al.).
- **Seasonal storage exists,** but it is a lake: Norway's reservoirs, reached from Denmark through a few GW of cable. That cable is today's first topic.

Closing line: the balance once more, with today's two terms marked.

The hourly balance, and what can move

Every hour, in *Energy System Models*' notation:

$$\underbrace{\gamma_h \bar{q}_w}_{\text{wind and solar}} + \underbrace{q_{g,h}}_{\text{dispatchable plants}} + \underbrace{g_h^d - g_h^c}_{\text{storage}} = \underbrace{D_h}_{\text{demand}} + \underbrace{x_h}_{\text{net exports}}$$

- **Trade**, x_h: a surplus is sold to the neighbours, a shortfall bought from them.
- **Flexible demand**, D_h: consumption moves to the hours when power is plentiful. Heat is the case, if it can wait.
- Today asks of both: which part of the problem does it relieve?



The hourly balance, and what can move

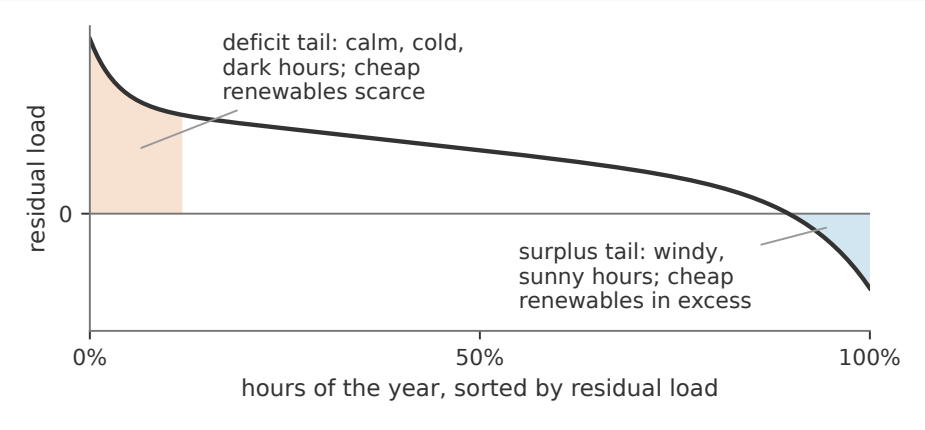
Lecture 5: Sector coupling and transmission · Where we left things

Lead line: two more of the four blue terms of lecture 4's balance, now in vermillion: the one that moves energy in space, and the one that moves demand in time.

- **The price λ_h** is the multiplier on this equation; with trade there is one equation, and one price, per zone.
- **Storage and trade are cousins:** storage moves a MWh from one hour to another, a wire from one place to another. Both are paid by price differences.
- **Flexible demand** is not a technology but a property of what the demand is for: heat has a store (the building, a tank) and, in district heating, a second fuel.

Closing line: "which part of the problem" needs a picture: the two tails.

The two tails of the balancing problem



Stylised: load minus wind and solar for every hour of the year, sorted from highest to lowest (lecture 4's residual load duration curve).

The two tails of the balancing problem

Lecture 5: Sector coupling and transmission · Where we left things



Lead line: the balancing problem has two tails, and each flexibility option can be asked which of them it relieves.

- **The deficit tail,** left: hours when little wind and sun meet high demand, typically calm, cold, dark weeks. Something else must produce: backup plants, imports, stored energy.
- **The surplus tail,** right: hours when wind and sun exceed demand, typically windy nights and sunny middays. The price falls to zero and the surplus is curtailed, unless it can be stored, used or exported.
- **Storage** works on both, moving energy from the right tail to the left; lecture 4 showed that its cost depends on how far apart in time the two tails are.

Closing line: trade first.

Trade over wires

Trade moves the surplus in space

The balance of lecture 4, one per zone z:

$$\sum_{g \in \mathcal{G}_z} q_{g,h} + \underbrace{\sum_{\ell:\, b_\ell = z} (1 - \phi_\ell) f_{\ell,h}}_{\text{imports, net of losses}} = D_{z,h} + \underbrace{\sum_{\ell:\, a_\ell = z} f_{\ell,h}}_{\text{exports}} \qquad (\lambda_{z,h})$$

- Every zone has its own balance, and its own price $\lambda_{z,h}$.
- A corridor ℓ carries a flow $f_{\ell,h}$ from zone a_ℓ to zone b_ℓ, and loses the share ϕ_ℓ of it on the way.
- Storage moves a surplus to another hour; trade moves it to another place, where residual demand is high.

Energy System Models, §6.3. Wind and solar are among the $q_{g,h}$ here.



Trade moves the surplus in space

Lecture 5: Sector coupling and transmission · Trade over wires

Lead line: trade adds a subscript to the balance, not a new theory: one market clearing per zone, joined by the flows.

- **Directed corridors:** every corridor appears twice, once for each direction, so every flow is nonnegative; losses are charged where the power arrives.
- **The capacity** of a corridor is what the market may trade on it, below what the wires could carry. Next slides.
- **Wind and solar** sit among the generators $q_{g,h}$ with their capacity factors in the upper bounds, as in ESM §3.

Closing line: a question before the wires.

Question

Can trade relieve the two tails of the balancing problem?

1. only the deficit tail
2. only the surplus tail
3. both
4. neither

Question

Lecture 5: Sector coupling and transmission · Trade over wires

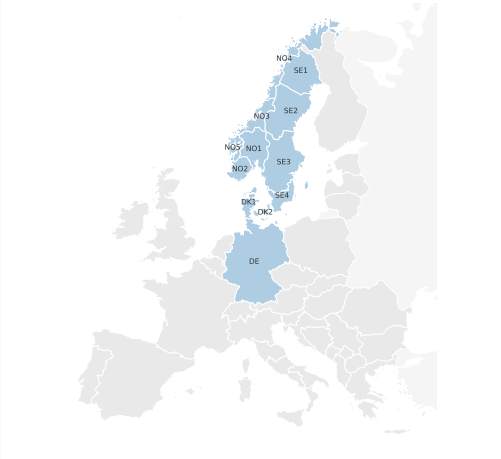


Lead line: the question the section answers: which tail does a wire relieve?

- **Answer 3, on two conditions.** Imports cover the deficit tail and exports drain the surplus tail, provided the neighbour's residual load differs from ours in those hours, and the wire has room.
- **The conditions** are the rest of the section: what a wire is to the model, what it is worth, and when neighbours differ.

Closing line: first, where the prices live: bidding zones.

Europe's market: bidding zones



Energy System Models, Fig. 6.1: the bidding zones of the day-ahead market; ESM's twelve highlighted.

Europe’s market: bidding zones

Lecture 5: Sector coupling and transmission · Trade over wires



Lead line: Europe’s market pretends there is no grid inside a zone: one price per zone, and transmission rationed only between zones.

- **Zones:** most are countries; Norway has five, Sweden four, Italy seven, Denmark two, DK1 for Jutland and Funen and DK2 for Zealand.
- **Congestion inside a zone** is still there, only hidden from the price. The system operator fixes it after the market by redispatch, and the cost ends in network tariffs; in Germany it runs into billions of euros a year.
- **Nodal pricing,** the American design, gives every node a price from a model of the whole grid, so internal congestion is priced rather than redispatched.
- **ESM’s network:** DK1, DK2, Germany, SE1–4 and NO1–5, solved for 2024.

Closing line: the wires between the zones, and the two kinds of current they carry.

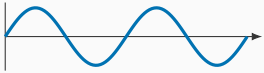
AC versus DC

DC, direct current



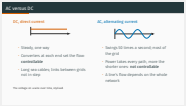
- Steady, one way
- Converters at each end set the flow: **controllable**
- Long sea cables; links between grids not in step

AC, alternating current



- Swings 50 times a second; most of the grid
- Power takes every path, more the shorter ones: **not controllable**
- A line's flow depends on the whole network

The voltage on a wire over time, stylised.



AC versus DC

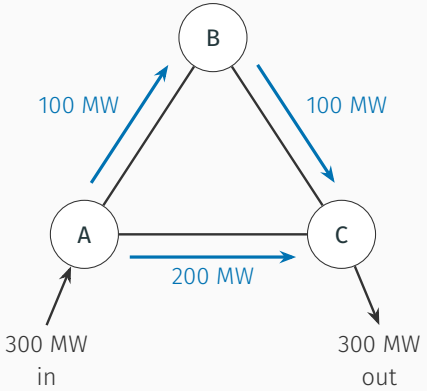
Lecture 5: Sector coupling and transmission · Trade over wires

Lead line: a DC line has a valve, its converters, so the operator chooses the flow; an AC network has none, and the flow on every line follows from where power enters and leaves.

- **Why the grid is AC:** transformers work only on AC, and high voltage cuts losses: the same power at twice the voltage takes half the current and loses a quarter as much. That is why AC won over DC in the 1890s.
- **One swing per area:** every generator in a synchronous area turns in step with the same 50 Hz swing; continental Europe is one area, the Nordics another. Only DC can join two of them, since its two ends need not swing in step.
- **Why DC under the sea:** an AC cable spends much of its capacity charging its own insulation, 50 times a second; a DC cable charges it once. The price is the converters.
- **What sets an AC flow:** the timing of the swing at each end of a line; a line carries power in proportion to the timing difference across it, divided by its reactance, which grows with its length (ESM §6.2).

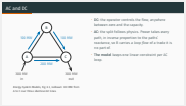
Closing line: what that means on a loop of three zones.

AC and DC



Energy System Models, Fig. 6.3, redrawn: 300 MW from A to C over three identical AC lines.

- **DC:** the operator controls the flow, anywhere between zero and the capacity.
- **AC:** the split follows physics. Power takes every path, in inverse proportion to the paths' reactance, so B carries a *loop flow* of a trade it is no part of.
- **The model** keeps one linear constraint per AC loop.



Lead line: on DC the operator sets the flow; on AC the flow splits over every path by physics, and the model needs one constraint per loop to say so.

- **The split:** the detour through B has twice the reactance of the direct line, so it carries half as much: 200 MW direct and 100 MW through B, a trade B is no part of.
- **Two consequences:** loop flows use capacity nobody sold, which is why an AC border's NTC is set low; and loading one AC line changes every flow on the loop. The continental case is trade between Germany and Austria that loops through Poland and the Czech Republic.
- **DC joins what AC cannot:** DK1 swings with the continent, DK2 with the Nordics, so every cable between them, and from them to the North, is DC.
- **Flow-based coupling:** since 2024 the Nordic day-ahead market puts the loop constraint into the market algorithm itself, rather than hiding it behind fixed NTCs.

Closing line: what an economist keeps of all this: four numbers.

A line, to an economist: four numbers

A line ℓ carries power from zone a_ℓ to zone b_ℓ; a line used both ways counts twice, once per direction.

- **Capacity** $\bar{f}_\ell$, MW: what the market may trade on it, the *net transfer capacity*, set below the thermal rating for security (DK1–Germany: 2.5 GW).
- **Losses** ϕ_ℓ: resistance turns flow into heat; ESM charges 3% per DC crossing.
- **Cost:** long lines are DC, and under the sea DC is the only option.
- **Control:** can an operator set the flow? On DC, yes; on AC, no.

AC overhead line	442	EUR/MW per km
DC submarine cable	1008	EUR/MW per km
DC converter station pair	166	thousand EUR/MW

Energy System Models, §6.2 and Table 3; a 40-year lifetime.



A line, to an economist: four numbers

Lead line: to the model a line is a capacity, a loss share, a cost, and one yes-or-no property that the physics decides.

- **NTC:** the market sees 2.5 GW on Jutland–Germany where the wires could carry several times that; the margin covers the failure of any single element and flows the market did not schedule.
- **Cost:** a DC line pays for its converters once, then for its cable per km. For a cable of the Skagerrak's class, 360 km under water, the table gives about 529 million EUR per GW; the European planners price the next Skagerrak cable at 850–1150 million EUR per GW.
- **Losses** rise with the square of the current, so a busy line loses proportionally more.

Closing line: the model, in one slide.

The model: zones and corridors

min	$\sum_h \sum_z \sum_{g \in \mathcal{G}_z} c_g \, q_{g,h}$	cost	
s.t.	$\sum_{g \in \mathcal{G}_z} q_{g,h} + \sum_{\ell:\, b_\ell = z} (1 - \phi_\ell) f_{\ell,h} = D_{z,h} + \sum_{\ell:\, a_\ell = z} f_{\ell,h}$	zonal balance	$(\lambda_{z,h})$
	$0 \le q_{g,h} \le \gamma_{g,h} \, \bar{q}_g$	plant capacity	$(\underline{\mu}_{g,h},\ \overline{\mu}_{g,h})$
	$0 \le f_{\ell,h} \le \bar{f}_\ell$	corridor capacity	$(\underline{\omega}_{\ell,h},\ \overline{\omega}_{\ell,h})$
	$\sum_{\ell \in C} \sigma_{C\ell} \, x_\ell \, f_{\ell,h} = 0$	each AC loop C	$(\zeta_{C,h})$

The model: zones and corridors

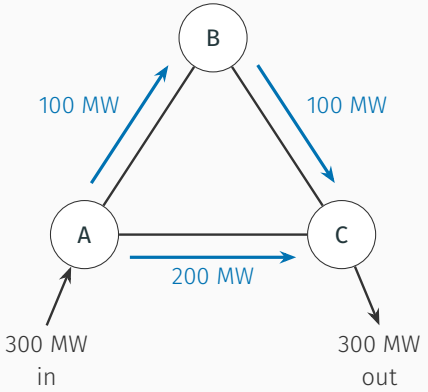


Lead line: lecture 3's dispatch model with the balance replicated per zone and two new constraints on the flows.

- **Nothing new in the plants:** each zone's generators keep their bounds and multipliers; the zone's price is the multiplier on its own balance.
- **The corridor capacity** is the NTC, with a multiplier pair like a plant's.
- **The loop constraint** holds only on AC loops; on a DC cable, and on an AC corridor that closes no loop (Germany–DK1), it is absent.
- **Nodal pricing** is the same problem with a node in place of a zone and the loop constraint on every loop of the full grid.

Closing line: the loop example of the previous slides, in these equations.

The loop flow, in the model's equations

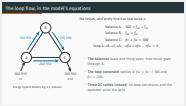


Energy System Models, Fig. 6.3, redrawn.

No losses, and every line has reactance x:

$$\begin{aligned}
\text{balance A:} \quad & 300 = f_{AB} + f_{AC} \\
\text{balance B:} \quad & f_{AB} = f_{BC} \\
\text{balance C:} \quad & f_{AC} + f_{BC} = 300 \\
\text{loop A}\to\text{B}\to\text{C}\to\text{A:} \quad & x f_{AB} + x f_{BC} - x f_{AC} = 0
\end{aligned}$$

- **The balances** leave one thing open: how much goes through B.
- **The loop constraint** settles it: $f_{AB} = f_{BC} = 100$ and $f_{AC} = 200$.
- **Three DC cables instead:** no loop constraint, and the operator picks the split.



Lead line: the loop example is the model on three zones: three balances and one loop constraint, and the loop constraint is what fixes the split.

- **The balances:** A sends 300 MW, B passes on what it receives, C takes 300 MW. Only two of the three are independent, so three flows are left with one degree of freedom.
- **The loop:** with $f_{AB} = f_{BC} = f$ and $f_{AC} = 300 - f$, the constraint reads $2xf = x(300 - f)$, so $f = 100$. The detour, twice as long electrically, carries half as much. The signs are the $\sigma_{C\ell}$: $+1$ on the lines run in the loop's direction, -1 on A$\to$C, run against it.
- **Capacity:** rate every line at 250 MW, and a transfer T puts $2T/3$ on the direct line, so the transfer stops at 375 MW, short of the 500 MW of wire touching C. A meshed grid can carry less than the sum of its lines.
- **With DC cables** any split satisfies the balances, and the operator could send all 300 MW direct.

Closing line: what the flow condition says about prices.

A DC cable: the flow condition

Differentiate the Lagrangian with respect to the flow on a DC cable from a to b:

$$(1 - \phi_\ell)\,\lambda_{b,h} - \lambda_{a,h} + \underline{\omega}_{\ell,h} - \overline{\omega}_{\ell,h} \;=\; 0$$

- **Nothing sent:** $\underline{\omega}_{\ell,h} = \lambda_{a,h} - (1 - \phi_\ell)\lambda_{b,h} \geq 0$. The far price, net of losses, does not cover the home price.
- **Used, below capacity:** $\lambda_{b,h} - \lambda_{a,h} = \phi_\ell\,\lambda_{b,h}$. The prices differ by the *loss wedge* and nothing else.
- **Congested:** $\overline{\omega}_{\ell,h} = (1 - \phi_\ell)\lambda_{b,h} - \lambda_{a,h} > 0$. The prices decouple; the spread net of losses is what one more MW of cable is worth in that hour.



A DC cable: the flow condition

Lead line: the flow condition is an arbitrage condition: power flows towards the costlier zone, and only while the spread covers the losses.

- **The flow appears** only in the two zonal balances and its own bounds: it costs nothing and closes no loop.
- **ESM's result on prices across a corridor:** across a slack corridor the prices differ by the loss wedge $\phi_\ell \lambda_{b,h}$; across a congested one, the spread net of losses is the shadow price of capacity, the $\overline{\mu}_g$ of lecture 3 attached to a wire instead of a plant.
- **On an AC corridor** the condition gains a term $\sum_C \zeta_{C,h} \sigma_{C\ell} x_\ell$ for every loop it lies on, so prices can differ across a slack AC line when a loop binds elsewhere: the price of loop flow.
- **Zonal prices** are then the planner's market clearing decentralised by geography, with a direct counterpart in the Nord Pool and EPEX area prices.

Closing line: so what is a cable worth?

What a MW of cable is worth

- **One more MW of corridor** lowers the year's cost by the sum of its hourly congestion prices (the envelope theorem):

$$\frac{\partial C^*}{\partial \bar{f}_\ell} = -\sum_h \overline{\omega}_{\ell,h}.$$

  Build it while $\sum_h \overline{\omega}_{\ell,h}$ exceeds the annuity of a MW of cable.

- **The rent the owner collects**, flow times spread, splits in two:

$$R_\ell = \sum_h f_{\ell,h}\left(\lambda_{b,h} - \lambda_{a,h}\right) = \underbrace{\sum_h \overline{\omega}_{\ell,h}\bar{f}_\ell}_{\text{scarcity rent}} + \underbrace{\sum_h \phi_\ell\, \lambda_{b,h} f_{\ell,h}}_{\text{value of the losses}}.$$

What a MW of cable is worth

Lecture 5: Sector coupling and transmission · Trade over wires

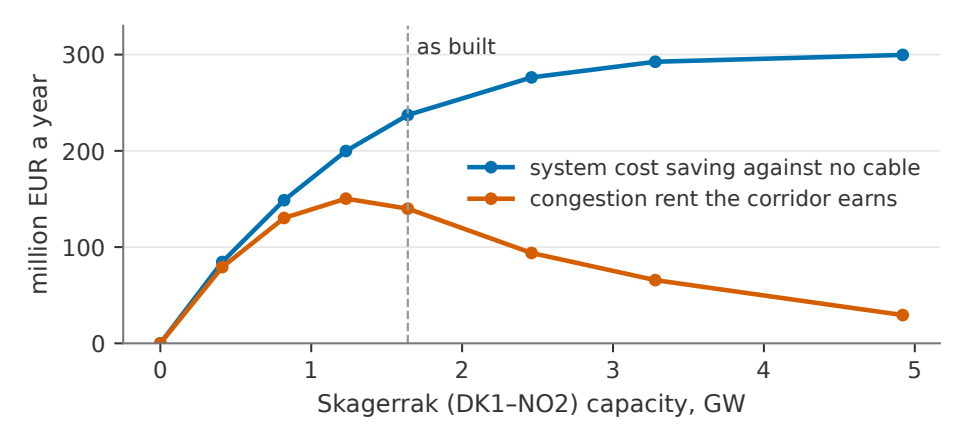


Lead line: a wire is paid, like a plant, by the hours its capacity binds.

- **Why the sum:** $\bar{f}_\ell$ enters the Lagrangian only through $\overline{\omega}_{\ell,h}(\bar{f}_\ell - f_{\ell,h})$ in each hour, so its derivative is $\sum_h \overline{\omega}_{\ell,h}$, and at the optimum that is what one more MW saves.
- **The split:** whenever the cable flows, the spread equals $\overline{\omega}_{\ell,h} + \phi_\ell \lambda_{b,h}$, and $\overline{\omega}_{\ell,h}$ is positive only when the flow is at $\bar{f}_\ell$.
- **The owners** are the system operators, who report the rent as *bottleneck income* and must by EU regulation spend it on more capacity or lower tariffs.
- **Decreasing returns:** each extra MW narrows the spread it earns, so the next MW is worth less than the last.

Closing line: the Skagerrak, swept.

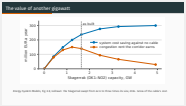
The value of another gigawatt



Energy System Models, Fig. 6.8, redrawn: the Skagerrak swept from zero to three times its size, 2024. Gross of the cable's cost.

The value of another gigawatt

Lecture 5: Sector coupling and transmission · Trade over wires



Lead line: the system's gain from a cable rises with its size, with decreasing returns; the cable owner's rent rises and then falls.

- **Social value:** the system cost saving against no cable, 237 million EUR a year as built and 300 at three times the size. Each GW closes more of the spread the next one would have earned.
- **Merchant value:** the rent peaks a little below today's size (140 as built) and falls to 29 at triple: a big enough cable competes away the spread it earns.
- **So a merchant investor** paid only the rent stops at the hump, while every GW beyond it still saves the system money: the argument for regulated transmission (Joskow and Tirole 2005).
- **Where the peak sits** is a property of this instance, its water and its fuel prices, not a finding.

Closing line: when does a cable pay at all?

When does trade pay?

- A wire earns the price spread, so trade pays only where prices differ.
- **In level:** different marginal costs, Nordic water against German gas and coal. Trade moves cheap energy to costly places.
- **In timing:** residual loads that do not move together, because the weather or the demand differs. Trade moves a surplus to where there is a deficit *in the same hour*.
- The balancing problem needs the second: a neighbour whose windless week is not ours.

When does trade pay?

Lecture 5: Sector coupling and transmission · Trade over wires

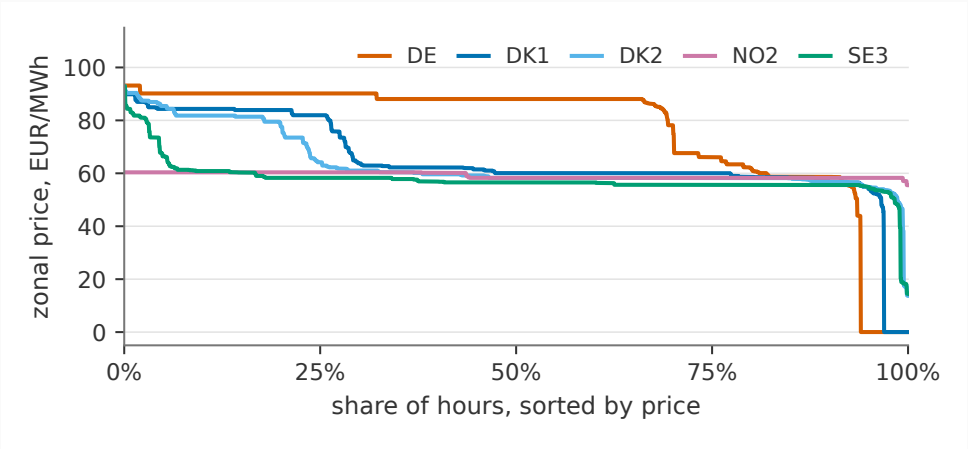


Lead line: trade pays when prices differ, and they differ in level or in timing; the two tails need the timing.

- **Level:** with a constant cost difference, a wire runs one way all year; it lowers costs but does nothing for the balance.
- **Timing:** with the same average costs but weather that differs, a wire runs both ways, exporting in our surplus hours and importing in our deficit hours. That is flexibility.
- **Denmark has both:** it sits between costly thermal Germany and cheap hydro Norway, and its wind differs from theirs.

Closing line: the prices, in ESM's model of 2024.

Denmark between two neighbours



Energy System Models, Fig. 6.4, redrawn: price duration curves of the five zones that face Denmark, the model of 2024.



Denmark between two neighbours

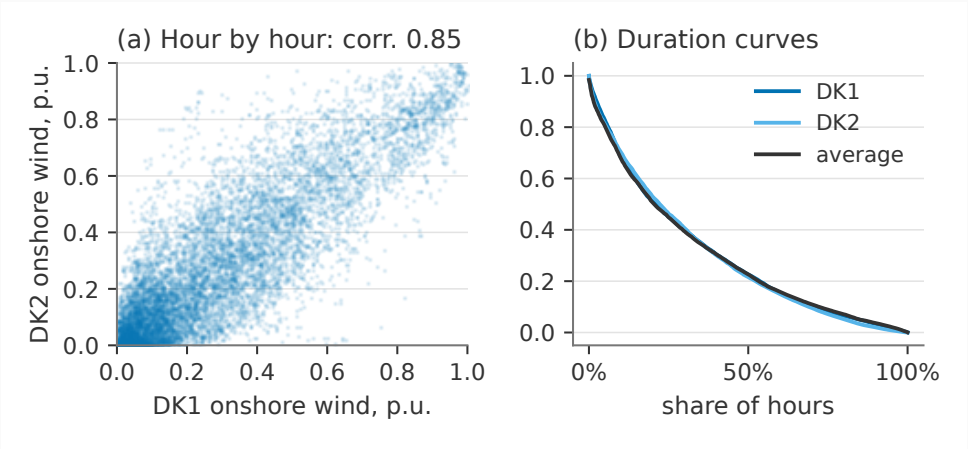
Lecture 5: Sector coupling and transmission · Trade over wires

Lead line: the Nordic prices are flat, Germany's swing, and Denmark takes a draw from each tail rather than averaging them.

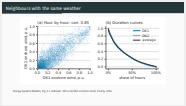
- **Flat north:** reservoir hydro is storage at national scale, and arbitrage flattens the price it trades on; the plateau is the water value (NO2 averages 59.2 EUR/MWh).
- **Germany** is thermal at the margin: gas plus carbon in the tight hours, coal below that, zero in wind surplus hours (average 76.9).
- **Denmark** gets the German price when the Nordic cables are full and the Nordic one when they have room (DK1 64.8, DK2 63.7). A small system whose price is mostly set abroad.

Closing line: what about Denmark's own two zones?

Neighbours with the same weather



Energy System Models, Fig. 6.7, redrawn: DK1 and DK2 onshore wind, hourly, 2024.



Neighbours with the same weather

Lecture 5: Sector coupling and transmission · Trade over wires

Lead line: at a correlation of 0.85, averaging the two Danish zones barely flattens the duration curve: Denmark cannot diversify its wind against itself.

- **Three hundred kilometres** is not far in weather terms; correlations fall roughly exponentially with distance, and Iberian and Nordic wind are close to uncorrelated.
- **So aggregation pays at distance:** the statistical case for a continental grid rather than national self-sufficiency.
- **Denmark's smoothing** runs through the price channel instead: Norwegian reservoirs take Danish surpluses, by leaving water in the lakes, and give it back in calm weeks.

Closing line: the answer to the question.

What trade does to the two tails

- **Both tails, when the neighbour differs:** imports cover the deficit tail, exports drain the surplus tail.
- **In space, not in time:** a calm week in Denmark is covered from full Norwegian lakes, which no four-hour battery can do.
- **Not when everyone is short at once:** a still, cold week over all of northern Europe, or a sunny noon everywhere.
- **Only up to the wire's size:** the hours it is full are the hours it earns, and the hours it cannot help more.



What trade does to the two tails

Lead line: trade relieves both tails, and by more than storage does, as long as the weather differs across the wire.

- **The asymmetry with storage:** storage needs the two tails to be close in time; trade needs them to be apart in space.
- **Its value to wind** comes mostly through the price channel, not through diversification among near-identical neighbours.
- **ESM §7** makes trade load-bearing: cut off from its neighbours, Denmark needs a much larger and costlier fleet for the same load (lecture 6).

Closing line: the gains are real; who gets them is another matter.

Who wins and who loses from a cable

	low-price, exporting zone	high-price, importing zone
price	rises	falls
consumers	lose	win
producers	win	lose
cable owner	collects the congestion rent	
net	positive: the system cost saving	

Energy System Models, Table 4.

The sum is the gain from trade. Two of the four cells are losses, and they are concentrated.



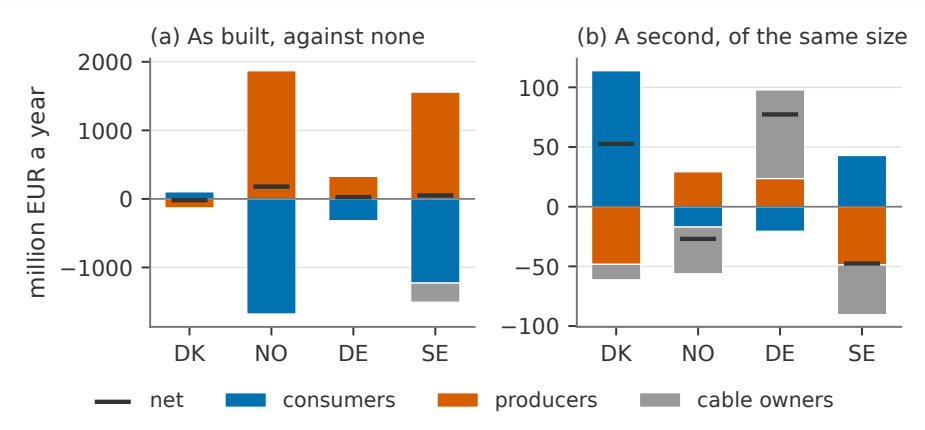
Who wins and who loses from a cable

Lead line: a cable joining a cheap zone to a costly one pulls the prices together, and that redistributes far more than it creates.

- **The exporter's consumers lose:** their cheap power is now sold abroad at a higher price. Norwegian households facing higher bills because their hydro can reach Germany is a recurring political controversy, and the model says the grievance is correct.
- **The country still gains,** because its producers gain more than its consumers lose: an argument for compensation, not for refusing to build.

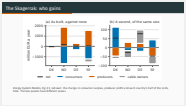
Closing line: ESM's numbers for the Skagerrak.

The Skagerrak: who gains



Energy System Models, Fig. 6.9, redrawn: the change in consumer surplus, producer profit and each country's half of the rents, 2024. The two panels have different scales.

The Skagerrak: who gains



Lead line: the transfers are an order of magnitude larger than the gain, and the next cable would have to be paid for by countries that do not gain from it.

- **As built, against none:** in wet 2024 Norway is the cheap side. Norwegian consumers pay 1682 million EUR more and producers earn 1869 more; Sweden moves with Norway. The system gains 237.
- **Denmark** is a small net loser (19) while owning half the cable: consumers gain, producers lose more.
- **A second cable:** Denmark gains 53 and Germany 77, while Norway, which would host and half-own it, loses 27, and Sweden loses too.

Closing line: so who pays for a cable?

Who pays for a cable?

- **Split it by ownership:** the Nordic convention, half each (Skagerrak: Energinet and Statnett). Simple, and unrelated to who gains.
- **Let the beneficiaries pay:** FERC's Order 1000 in the US; in the EU, cross-border cost allocation for projects of common interest, decided by the regulators or by ACER.
- **Let the cable pay for itself:** a merchant line keeps its rent, and stops at the hump.
- **Bargain over the surplus:** cooperative game theory, such as the Shapley value.

Energy System Models, §6.7. Hogan (2018) is the primer; Joskow and Tirole (2005) on merchant lines.



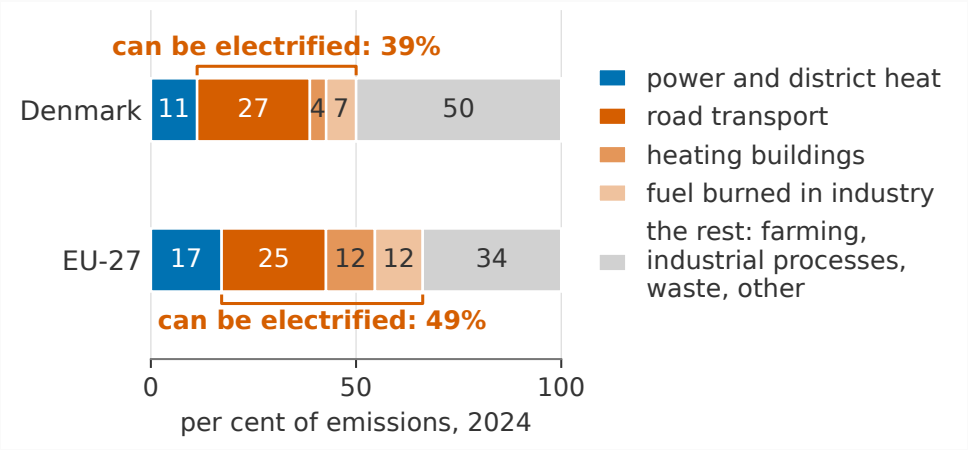
Lead line: the benefit is spread over consumers, producers and owners in several countries; the cost has to land on somebody's tariff.

- **By ownership:** on the Skagerrak figure, the two halves of the rent are small and alike while the two countries' nets differ by some two hundred million EUR.
- **By benefit:** the allocated benefit includes third countries, which is what makes it contentious; the benefits are hard to measure and easy to dispute (Hogan 2018).
- **Merchant:** too small a cable, and reasons to build it in the wrong place and keep it scarce; EU rules treat it as an exemption.
- **The rent** is the money on the table: by EU regulation it may only fund capacity or lower tariffs.

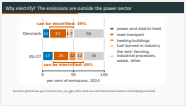
Closing line: the second new margin: demand, electrified.

Electrification and heat

Why electrify? The emissions are outside the power sector



Eurostat, greenhouse-gas inventory (env_air_gge), 2024; land use and international aviation and shipping excluded.



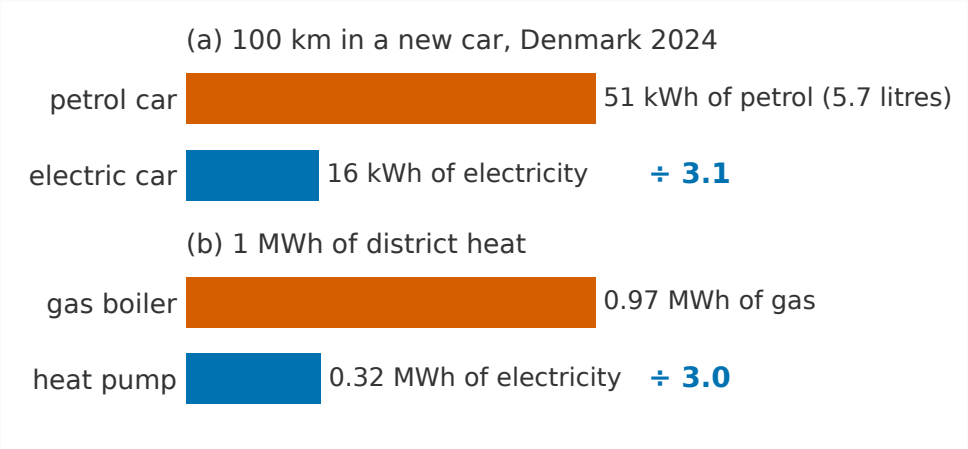
Why electrify? The emissions are outside the power sector

Lead line: most emissions come from fuel burned outside the power sector, in engines, boilers and furnaces that electricity can take over; that is the potential of electrification.

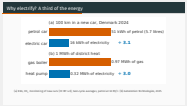
- **Three times the power sector:** road transport, heating buildings and industry's fuel emit about three times what power and district heat do: 39% against 11% in Denmark, 49% against 17% in the EU.
- **Danish power is mostly clean already:** wind and biomass (which the inventory counts at zero) have largely replaced coal; power and district heat emitted 4 Mt in 2024, against 25 Mt in 1990.
- **Only parts of industry:** low-temperature process heat suits a heat pump; high-temperature furnaces gain less from electricity, and some will need hydrogen or carbon capture.
- **Out of reach:** farming (animals, manure and soils: 31% of Danish emissions) and the chemistry of cement burn no fuel, so electricity cannot cut their emissions.

Closing line: why electricity can do the job cheaply: it needs a third of the energy.

Why electrify? A third of the energy



(a) EEA, CO_2 monitoring of new cars (CC BY 4.0), test-cycle averages; petrol at 32 MJ/l. (b) *Generation Technologies*, 2025.



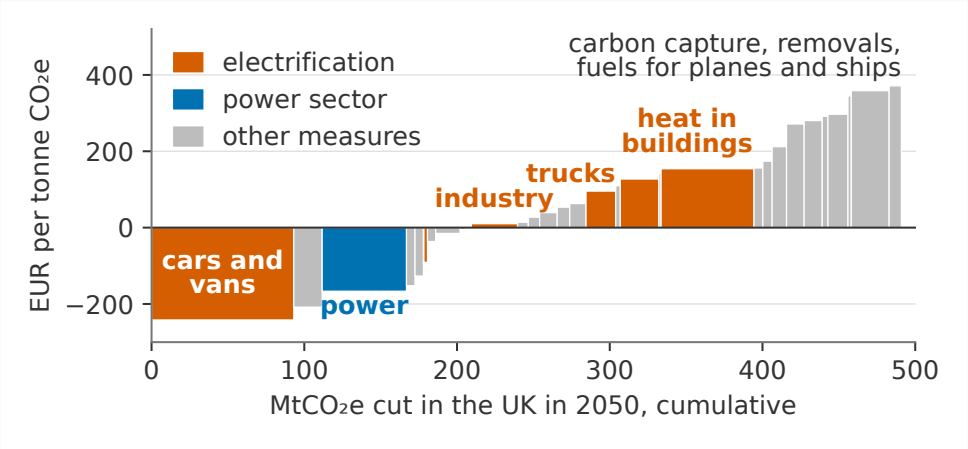
Why electrify? A third of the energy

Lead line: an electric car or a heat pump does the same job on about a third of the energy of the fuel it replaces: a motor wastes far less than an engine, and a pump moves heat rather than making it.

- **The engine wastes heat:** most of the energy in petrol leaves a car's engine as heat; an electric motor turns most of its electricity into motion.
- **The heat pump moves heat:** it lifts heat from the outdoor air, so a MWh of electricity delivers about three of heat; a boiler turns a MWh of gas into about one.
- **In tonnes:** a clean MWh saves 0.7 t of CO_2 in an electric car, 0.6 t in a heat pump and 0.3 t replacing gas-fired power; only replacing coal saves more (0.9 t; UK Climate Change Committee, Fig. 10.8).
- **A cautious comparison:** the new electric cars weigh 2.1 t on average, the petrol cars 1.3 t; between cars of the same size the gap is wider.

Closing line: energy saved is money saved: where that puts electrification on a cost curve.

Why electrify? The cheap end of the cost curve



Climate Change Committee (2025), *The Seventh Carbon Budget*, Fig. 4.6, redrawn; Open Government Licence v3.0; 1.15 EUR/GBP.



Why electrify? The cheap end of the cost curve

Lead line: on a cost curve for a whole economy, electrification supplies about half the tonnes and none of the costly ones; the costliest tonnes need carbon capture, removal from the air, or new fuels.

- **Reading the curve:** each block is a group of measures, its width the tonnes it cuts in 2050, its height the average cost of a tonne net of the energy saved (2023 prices); below zero, it saves money.
- **Cars save money:** electric cars and vans, the largest block, save about 240 EUR/t: costly to buy, cheap to run. In 2025 they still cost 250 EUR/t.
- **Heat is mid-curve:** about 150 EUR/t in British homes. A pump is costly to install and cuts running costs only if its COP exceeds the ratio of the power price to the gas price (about 4 there today).
- **The costly end:** carbon capture, removals and new fuels for planes and ships cost 210 to 370 EUR/t: costly plant, and hungry for energy.
- **A curve for 2050:** it assumes the Committee's clean power system; cleaning up power, the blue block, saves money too.

Closing line: with fossil power at the margin, the tonnes a heat pump avoids shrink and its cost per tonne rises.

Why electrify? A heat pump moves heat

Per MWh of district heat, 2025 costs:

	gas boiler	heat pump
energy in	about 1 MWh of gas	about 0.3 MWh of electricity (COP 3.15)
cost, EUR	37 + carbon (53 at 80 EUR/t)	50 (electricity at 60 EUR/MWh)
CO_2, t	0.19	0 with clean power; 0.11 with gas; 0.30 with coal

- Switching costs 68 EUR per tonne avoided if the electricity is clean, 161 if it comes from gas.
- With coal at the margin, the heat pump emits *more* than the boiler.

Generation Technologies: district-scale heat pump and gas boiler, both at the heat pump's 35% utilisation.


Why electrify? A heat pump moves heat

Per MWh of district heat, 2025 costs.

	gas boiler	heat pump
energy in	almost 1 MWh of gas	almost 0.3 MWh of electricity (COP ≈ 3.5)
cost, DKK	81 = carbon 0 + gas 81	53 (electricity at 182 DKK/MWh)
CO_2, t	0.19	0 with clean power, 0.11 with gas, 0.20 with coal

Switching costs 60 EUR per tonne avoided if the electricity is clean, 161 if it comes from gas.

With coal at the margin, the heat pump emits more than the boiler.

Illustrative technologies, electricity price, heat pump and gas boiler, tenth of the heat sector's 19% of emissions.

Why electrify? A heat pump moves heat

Lecture 5: Sector coupling and transmission · Electrification and heat

Lead line: a heat pump turns one MWh of electricity into about three of heat, which makes electrified heat cheap abatement, but only as clean as the power behind it.

- **Moving, not making:** a heat pump moves heat from outdoors to indoors, which is why its COP exceeds one; a boiler converts, and stays near one.
- **The abatement cost:** the heat pump's extra cost over the boiler without carbon, divided by the tonnes avoided. At the carbon price the technology note assumes, the heat pump is already the cheaper of the two.
- **The marginal plant decides:** the right emissions number is the marginal intensity of the electricity, not the average; ESM §5.3's rollout finds coal at the margin in a sixth of Danish hours, and a heat pump rollout that cuts little without a carbon price.

Closing line: the second motivation is the balance itself.

Flexible demand: a margin, or a new load?

- **A margin:** heat is cheap to store, in a tank, in the pipes, in a building's walls; and district heating can switch to a boiler when power is costly. So electrified heat could move to the hours when power is plentiful.
- **A new load:** heat demand is driven by the weather. It peaks in the cold hours, when a heat pump's COP is lowest and the wind may be calm.
- Which it is depends on what stands behind the heat pump: a store, a second fuel, or nothing.



Flexible demand: a margin, or a new load?

Lead line: electrification may make electricity demand more flexible, but that is not certain a priori: it adds a large, peaked, weather-driven load.

- **ESM §5's heat pump** has a gas boiler behind it, so it switches off when power is costly: the note's first price-responsive consumer.
- **Flexibility follows the appliance:** Danish smart meter data show households responding to hourly prices mainly when they own something with a store or an alternative, an electric car above all (ESM §5.2).
- **The workshop paper** studies individual heat pumps with no second fuel, with and without a heat store.

Closing line: the same question as for trade.

Question

Can flexible demand relieve the two tails of the balancing problem?

1. only the deficit tail
2. only the surplus tail
3. both
4. neither



Question

Lecture 5: Sector coupling and transmission · Electrification and heat

Lead line: the answer depends on whether the demand was already there: existing demand made flexible relieves both tails, while a new load such as electrified heat relieves the surplus tail at best.

- **Answer 3 for existing demand made flexible:** consumption moved out of the scarce hours shrinks the deficit tail, and moved into the cheap hours it shrinks the surplus tail.
- **At best answer 2 for a new load:** it only adds consumption, so it can fill in the surplus tail but never brings the deficit tail below where it was without the load. On a cold, calm evening it has nothing to supply.
- **What stands behind it decides how close it gets:** with a boiler or a long store it stays out of the scarce hours and leaves the deficit tail as it was (ESM §5's heat pump); a store of a few hours avoids short scarcities but not a cold, calm week; with nothing behind it, it follows the heat load and adds to the deficit tail.

Closing line: the workshop paper studies the last two cases. First, how heat markets differ from power markets.

Heat markets are not power markets

	electricity	district heat
reach	a continent, by wire	a town, by pipe
trade	across borders, hour by hour	none beyond the network
storage	costly (lecture 4)	cheap: tanks, pipes, buildings
balance	every second	with hours of slack
price	hourly, from a market	a tariff, from the heat company's costs

The heat pump joins the two: a consumer in one market, a producer in the other.



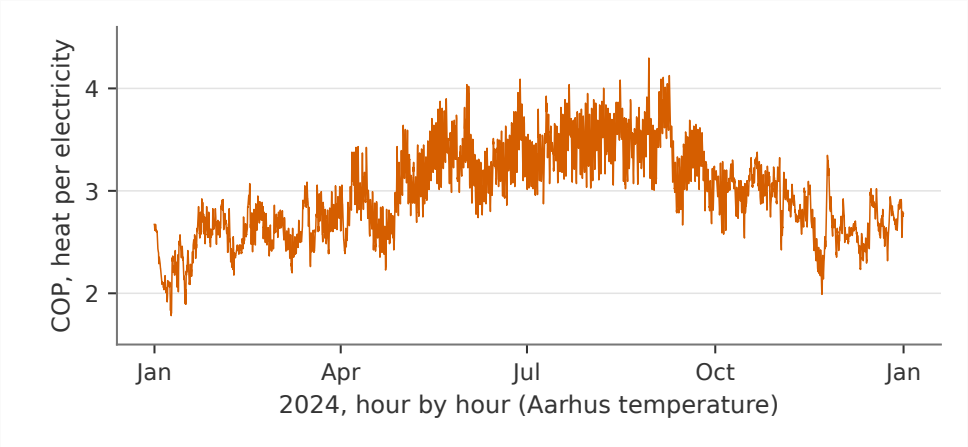
Heat markets are not power markets

Lead line: heat is local and easy to store, electricity continental and hard to store; the heat pump couples a sluggish market to a fast one.

- **Why no trade:** hot water loses heat and costs pumping over distance, so a heat market ends where its pipes end.
- **Why little balancing trouble:** the water in the pipes and the mass of the buildings hold hours of heat, so a heat system need not match supply to demand every second.
- **The price:** in ESM §5, the heat price is set by whichever technology is marginal, the boiler's cost or the electricity price divided by the COP; in reality, district heating tariffs are regulated and based on costs.

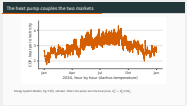
Closing line: the coupling, and its catch.

The heat pump couples the two markets



Energy System Models, Fig. 5.1(b), redrawn. When the pump sets the heat price, $\lambda_h^H = \lambda_h^E / COP_h$.

The heat pump couples the two markets



Lead line: the coefficient of performance is the exchange rate between the two markets, and it is worst exactly when heat is most needed.

- **The COP** averages 3.0 over 2024, from 1.8 in the coldest hours to 4.3 in summer: the colder the air, the more work it takes to lift heat to the network's temperature.
- **The adverse correlation** is the signature of electrified heat: a load that is largest when the COP is lowest arrives on top of a power system whose tight hours are the cold, dark, calm ones.
- **The coupling:** the pump draws q_h^E from the power market and delivers $COP_h \, q_h^E$ to the heat market, so when it is marginal the two prices are linked through the COP.

Closing line: does all this help integrate wind? The workshop.

Workshop: Ruhnau et al. (2020)

Today builds on

***Energy System Models*, section 5:** the heat market and the heat pump, and the rollout (§5.3); and §3.4 for the capture price and the value factor.

Ruhnau et al. (2020), *Heating with wind: Economics of heat pumps and variable renewables*. Sections 1–5.

Optional, on trade: Gonzales et al. (2023).

The question

The hope (the paper's p. 2): heat demand and wind are both higher in winter, and heat is cheap to store, so heat pumps should use wind "over-proportionately" and help integrate it.

heat pumps from 0 to 15% of consumption: their load cost	+21 EUR/MWh
30% wind, 15% heat pumps, wind capacity held: wind's value	+2 to 3 EUR/MWh
30% wind, 15% heat pumps, wind share held: wind's value	about 0

Why so much for the heat pumps, and so little for wind?



The question

Lecture 5: Sector coupling and transmission · Workshop: Ruhnau et al. (2020)

Lead line: the paper's answer is sober on both counts: heat pumps cost the power system more the more there are, and they do little for wind once wind grows with them.

- **The model:** EMMA, a linear investment and dispatch model of five north-western European markets, which Hirth used for lecture 3's paper.
- **The load cost** is the paper's new metric, defined on the next slide.
- **The paper's own figures** are not reprinted: the paper is not openly licensed. The solutions redraw them schematically.

Closing line: part 1.

Part 1

Setup: two prices, weighted two ways

$$\text{wind's market value (4)} = \frac{\sum_h w_h \lambda_h}{\sum_h w_h} \qquad \text{heat pump load cost (5)} = \frac{\sum_h l_h \lambda_h}{\sum_h l_h}$$

- λ_h the price, w_h wind's output, l_h the heat pumps' consumption, in hour h.
- **The model:** the planner builds everything except the wind turbines and the heat pumps, whose shares the authors set, and it can shed load.
- **In §3.1–3.2** the heat pumps follow the heat load, whatever the price.
- Keep the paper open at its figures: part 1 reads Figs. 1–7.

Setup: two prices, weighted two ways



Lead line: the paper's device is two averages of the same prices, one weighted by a supply, the other by a demand.

- **Wind's market value** is the capture price of ESM §3.4; the load cost is its mirror, what a MWh of variable demand costs the system.
- **The paper sums over five countries** as well as hours; the slide drops the country index.

Closing line: thirty minutes, three questions.

What heat pumps cost, and what they do for wind

1. The research question: why it matters; the two opposing forces on wind's value.
2. Heat pumps alone: Fig. 1; new capacity (Fig. 2); load shedding (Figs. 2–3).
3. Heat pumps and wind: Fig. 5's two patterns; the two effects in Figs. 6–7.

Stop point

Why the load cost rises steeply as heat pumps grow; heat pumps raise wind's value at a fixed wind capacity, not at a fixed wind share.

The full questions, the hints and the stretch are on the worksheet.

Solutions 1

Solutions 1, Q1–Q2: the questions

Q1. Why the research questions matter; the two opposing forces that make the effect on wind's value "ambiguous ex ante".

Q2. Fig. 1; what capacity the system adds for heat pumps (Fig. 2); load shedding, and why its cost barely matters (Fig. 3).

Solutions 1, Q1: why it matters, and two forces

- **Q1. Why it matters:** heat pumps decarbonise heat, wind decarbonises power, and they meet in the power market. Do they help or hinder each other there?
- **Q1. Force one:** heat pumps add demand, more of it in the windy season, which lifts the prices wind sells at.
- **Q1. Force two:** the wind share is held, heat pumps included, so more demand means more wind, which lowers its value again.
- **Q1. Hence** a numerical model: theory cannot say which force wins. Fig. 7 separates them.

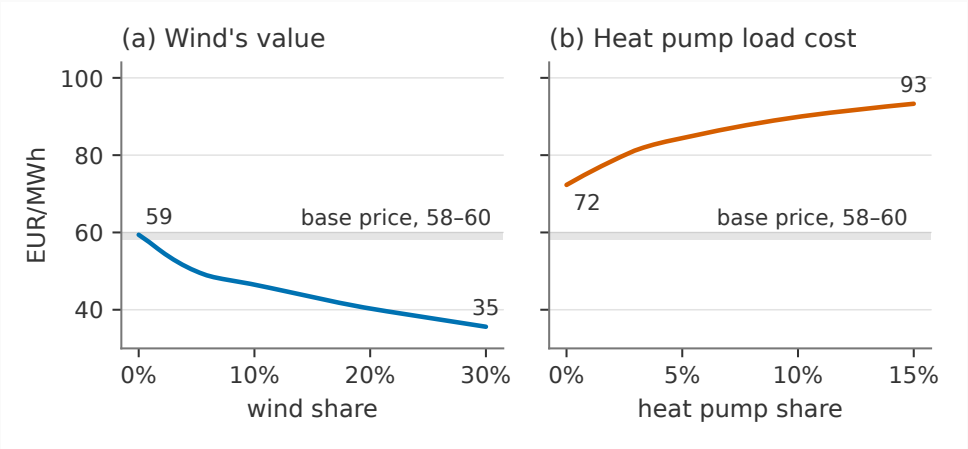


Lead line: electrifying heat eases the power system's own transition only if heat pumps are cheap to serve and help wind; whether they help wind is ambiguous.

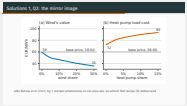
- **The research questions (p. 2):** how the load cost evolves with heat pump expansion; the effect of heat pumps on wind's value, and of wind on the load cost; the effect of heat pump technology and thermal storage on both.
- **Why hold the share:** renewable targets are written as shares, so electrifying heat must go with more renewables (p. 2).
- **Beyond the question, which force wins:** with the share set by policy, the shape of the heat pump load decides. A load shaped like the rest of demand would scale the whole system up and leave wind's value where it was; it rises only as far as heat pumps consume in windy hours more than the rest of demand. Were wind built on market terms instead, new wind would stop where its value is back at its cost (p. 12).

Closing line: Fig. 1.

Solutions 1, Q2: the mirror image



After Ruhnau et al. (2020), Fig. 1, redrawn schematically on one price axis: (a) without heat pumps, (b) without wind.



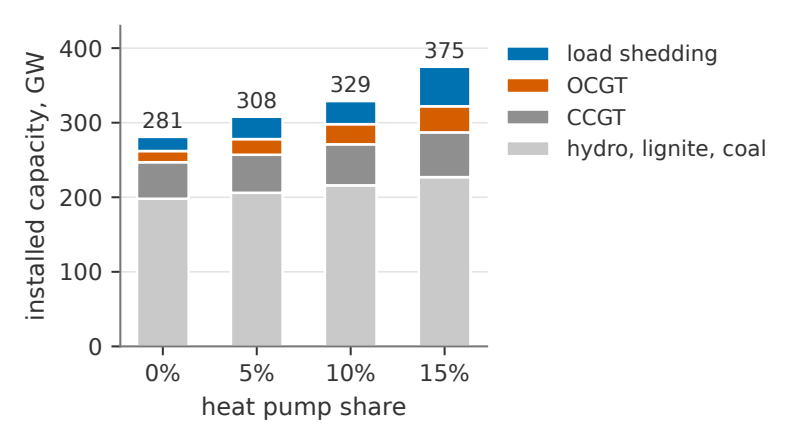
Solutions 1, Q2: the mirror image

Lead line: wind's value falls below a steady base price as wind grows, and the heat pumps' load cost rises above it as they grow.

- **Wind:** lecture 3's cannibalisation; its output arrives all at once and depresses the price of its own hours (40% of its value lost at 30% wind, p. 5).
- **Heat pumps:** the same mechanism reversed. Inflexible heat pumps all consume in the same cold hours and push up their prices (29% more at 15%, p. 5).
- **The base price** stays put: in the long run it equals the levelised cost of the baseload plant that runs every hour (footnote 10).

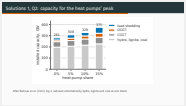
Closing line: what the system builds for the heat pumps.

Solutions 1, Q2: capacity for the heat pumps' peak



After Ruhnau et al. (2020), Fig. 2, redrawn schematically; hydro, lignite and coal as one block.

Solutions 1, Q2: capacity for the heat pumps' peak



Question

The value of lost load doubles, from 1,000 to 2,000 EUR/MWh. What happens to the heat pumps' load cost?

1. its scarcity part roughly doubles
2. it barely moves
3. it falls



Question

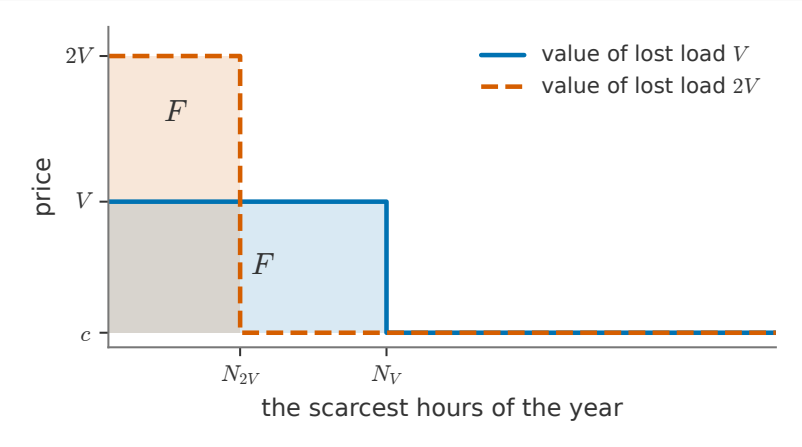
Lecture 5: Sector coupling and transmission · Solutions 1

Lead line: the question behind Fig. 3 (right): does the load cost depend on how high scarcity prices go?

- **Answer 2.** A higher value of lost load makes more peaking plant worth building, so the number of scarcity hours falls in inverse proportion to the scarcity price (p. 6), and the scarcity rent, what the heat pumps pay through it, stays the same. Next slide.

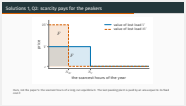
Closing line: why the hours adjust.

Solutions 1, Q2: scarcity pays for the peakers



Ours, not the paper's: the scarcest hours of a long-run equilibrium. The last peaking plant is paid by an area equal to its fixed cost *F*.

Solutions 1, Q2: scarcity pays for the peakers



Lead line: in a long-run equilibrium the scarcity prices pay for the peaking plant, so what they add up to is the plant's fixed cost, whatever the price cap.

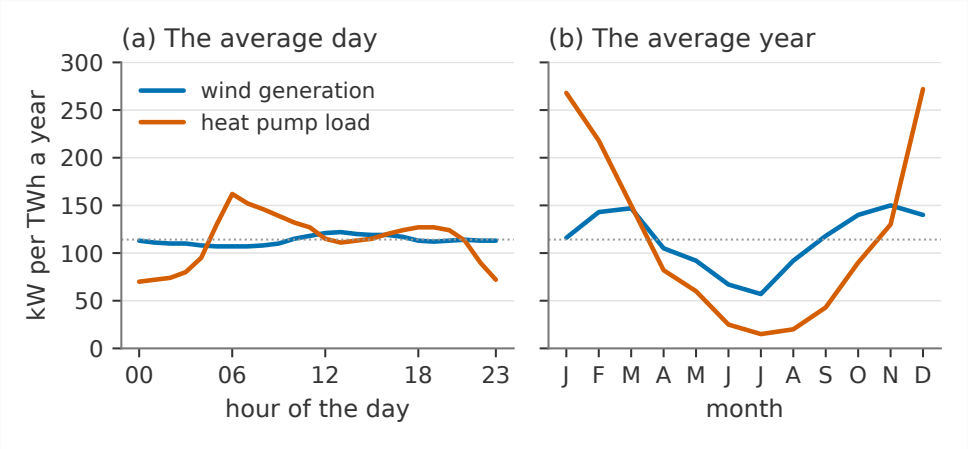
- **Zero profit:** the last peaker is built until $N(V - c) = F$: N scarcity hours at price V, running cost c, fixed cost F per MW and year. Double V and N roughly halves.
- **So the load cost measures a cost:** scarcity prices are how the market charges the heat pumps for the peak capacity they need. Lecture 6 takes up the zero-profit condition in full (ESM §7).
- **The stretch, weather years (Fig. 4):** the load cost varies far more between years than wind's value does; a peaky year, with few full load hours, costs more, unless its peaks miss the system's, as in 2008 (footnote 12).

Closing line: now heat pumps and wind together.

Solutions 1, Q3: the questions

Q3. Fig. 5: the daily and seasonal patterns, and a correlation of 0.11. Figs. 6–7: what heat pumps do to wind's value; what the dashed line holds fixed, and the two effects it separates.

Solutions 1, Q3: winter, but not the same hours



After Ruhnau et al. (2020), Fig. 5, redrawn schematically: kW per TWh of annual energy; dotted, a flat profile.



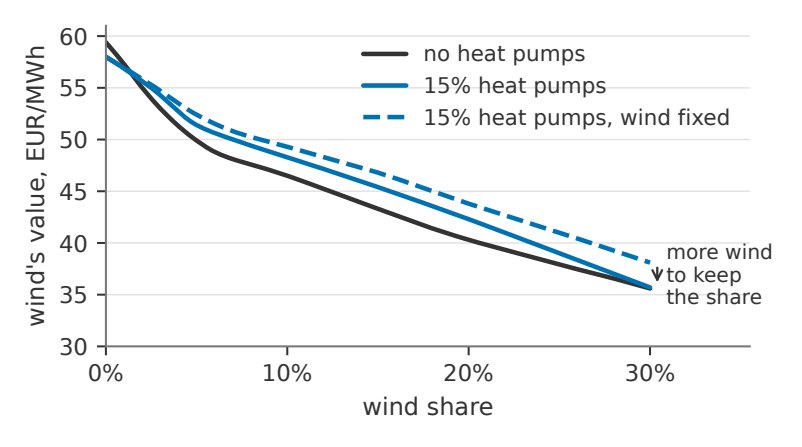
Solutions 1, Q3: winter, but not the same hours

Lead line: wind and heat pumps line up by season, not by hour.

- **The year:** both are higher in winter, the heat pumps much more so. **The day:** wind is almost flat; the heat pumps peak in the morning.
- **Correlation** 0.11**:** weakly positive. Within a season the hours hardly match: heat pumps run as hard in a cold, calm week as in a cold, windy one.
- **Fig. 6:** wind lowers the load cost clearly, and less as heat pumps grow; heat pumps barely move wind's value, and not at all at 30% wind.

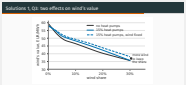
Closing line: why nothing at 30%? Fig. 7.

Solutions 1, Q3: two effects on wind's value



After Ruhnau et al. (2020), Fig. 7, redrawn schematically. Dashed: the wind capacity of no heat pumps.

Solutions 1, Q3: two effects on wind's value



Lead line: at a fixed wind capacity heat pumps raise wind's value; at a fixed wind share the extra wind takes it back.

- **The dashed line** holds the amount of wind: only demand grows, more of it in the windy season, and wind's value rises (by 7% at 30% wind): Q1's first force.
- **The solid line** holds the share: wind's output grows with consumption and cannibalises its own value: the second force.
- **Which wins:** the first at 5–20% wind; at 30% they balance out (p. 7).

Closing line: part 2: what brings the load cost down.

Part 2

Setup: what takes the cost back

- **The question of §3.3–5:** can the load cost increase of part 1 be brought down, and what does that do for wind?
- **Two remedies** besides wind: “system-friendly” heat pumps (§3.3, Table 1) and thermal storage (§3.4, Eqs. (9)–(10)).
- **Other flexibility** the planner builds: interconnectors and pumped hydro.
- Part 2 reads Figs. 8–13.

Part 2 is self-contained: what it needs from part 1 is at its top on the worksheet.



Setup: what takes the cost back

Lecture 5: Sector coupling and transmission · Part 2

Lead line: each remedy is judged twice, by what it does to the load cost and by what it does for wind.

- **Everything else is still built by the planner,** including the other options for flexibility, which is what question 2 turns on.

Closing line: three questions.

What takes the cost back, and what it does for wind

1. System-friendly heat pumps: what the term means; the load cost (Fig. 8), and why little for wind.
2. Flexible heat pumps: what flexibilization means; the load cost (Fig. 9), and why little for wind (Figs. 10–12).
3. What it adds up to: Fig. 13; the answer to the title question.

Stop point

Both remedies lower the load cost and, with wind, more than offset its increase; neither does much for wind's value.

The full questions, the hints and the stretch are on the worksheet.

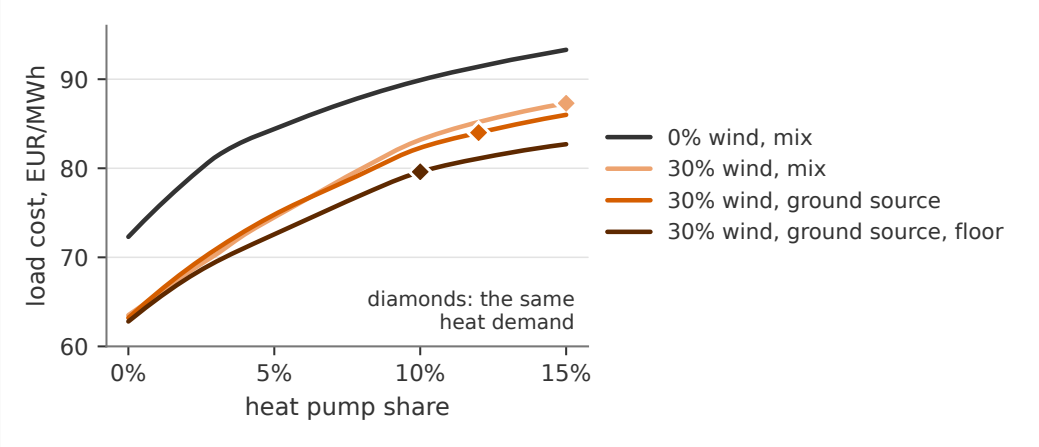
Solutions 2

Solutions 2, Q1–Q2: the questions

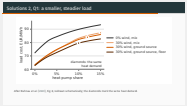
Q1. What “system-friendly” means; what it does to the load cost (Fig. 8), and why it does little for wind.

Q2. What “flexibilization” means; what it does to the load cost (Fig. 9), and why it does little for wind (Figs. 10–12).

Solutions 2, Q1: a smaller, steadier load



After Ruhnau et al. (2020), Fig. 8, redrawn schematically; the diamonds mark the same heat demand.



Solutions 2, Q1: a smaller, steadier load

Lead line: a system-friendly heat pump draws a smaller and steadier load for the same heat, and is cheaper to serve on both counts.

- **What it means:** ground source with floor heating works across a smaller, steadier temperature difference than air source with radiators, so its COP is higher and steadier (footnote 13).
- **Profile:** at the same share, a steadier load stays out of the scarcest hours. **Volume:** the same heat takes a smaller share, further down the curve (the diamonds). Together about 8 EUR/MWh (p. 8).

Closing line: and for wind?

Solutions 2, Q1: why so little for wind

- **Q1. What a steadier load avoids** is its peaks: cold hours with scarcity prices.
- **Q1. Those hours are mostly calm:** wind produces little in them.
- **Q1. In the windy hours** where wind earns its value, the heat pump load hardly changes, so neither does wind's value ("no significant implications", p. 8).

Solutions 2, Q1: why so little for wind

Lecture 5: Sector coupling and transmission · Solutions 2

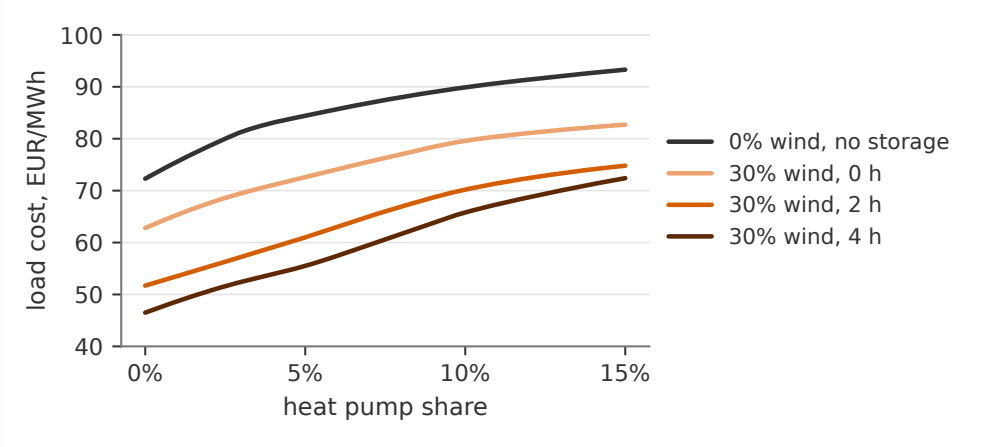


Lead line: what makes heat pumps cheaper to serve happens in hours in which wind sells little.

- **The paper states the result** without the reason; the reason is the one the worksheet's hint points to.

Closing line: the second remedy, storage.

Solutions 2, Q2: storage that holds hours



After Ruhnau et al. (2020), Fig. 9, redrawn schematically: ground source with floor heating at 30% wind.

Solutions 2, Q2: storage that holds hours

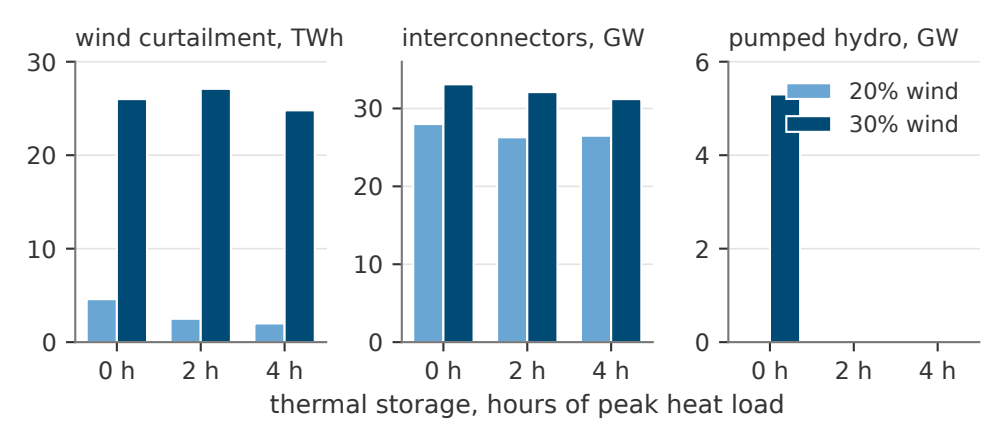


Lead line: flexible heat pumps move their demand with prices, but only by a few hours; that lowers the load cost a great deal, and less the more heat pumps there are.

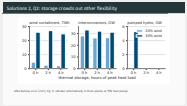
- **What flexibilization means:** with thermal storage the heat pumps no longer follow the heat load; they consume in cheap hours and draw on the store later. The store holds two or four hours of peak heat load, and heat leaks out of it.
- **Why less at high shares (p. 10):** the heat pumps' own load then drives prices by season, and a store of hours cannot move winter into summer.
- **The stretch:** with 5% lost on the way in and 1% an hour, 0.95×0.99^{k} is left after k hours: most of it after four hours, a quarter lost after a day, almost all after a week.

Closing line: and for wind?

Solutions 2, Q2: storage crowds out other flexibility



After Ruhnau et al. (2020), Fig. 11, redrawn schematically in three panels, at 15% heat pumps.



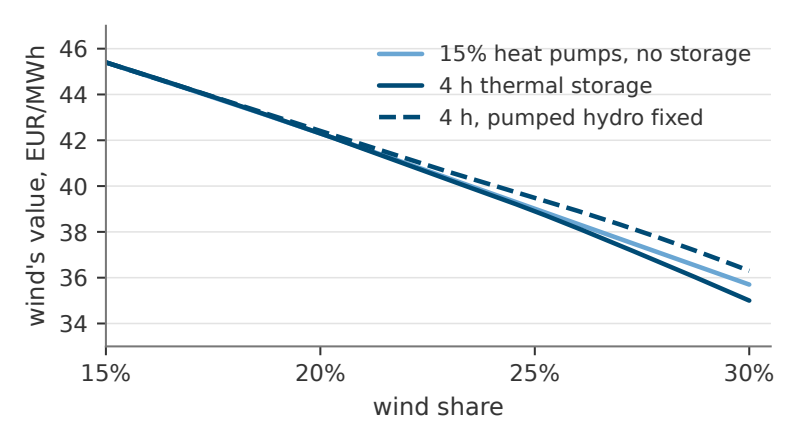
Solutions 2, Q2: storage crowds out other flexibility

Lead line: thermal storage flattens the price spreads that pay for interconnectors and pumped hydro, and fewer of them are built.

- **Fig. 10:** storage does nothing for wind's value, and at 30% wind lowers it slightly; counterintuitive, since the heat pumps do move load towards windy hours (curtailment falls at 20% wind).
- **At** 30% **wind** the planner builds fewer interconnectors and no pumped hydro: "thermal storage tends to reduce price volatility, which at the same time, is the driver for investing in flexibility" (p. 10).

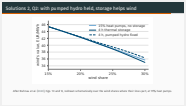
Closing line: what if the pumped hydro stays?

Solutions 2, Q2: with pumped hydro held, storage helps wind



After Ruhnau et al. (2020), Figs. 10 and 12, redrawn schematically over the wind shares where their lines part, at 15% heat pumps.

Solutions 2, Q2: with pumped hydro held, storage helps wind



Lead line: thermal storage helps wind on top of flexibility that stays, not in place of flexibility that goes.

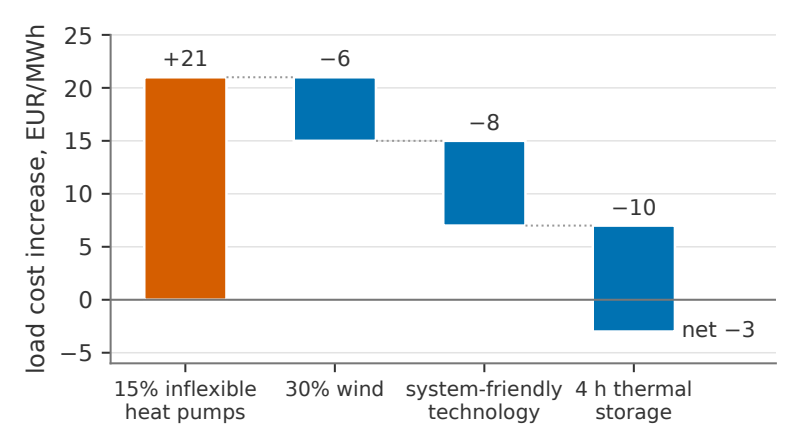
- **Fig. 12** holds pumped hydro at its level without thermal storage, and then the storage raises wind's value (around 1 EUR/MWh at 30% wind, p. 10).
- **So Fig. 10 is a substitution,** not a failure of thermal storage: flexibility options compete rather than add up (p. 12).
- **Beyond the question, lecture 4:** in Zerrahn et al. (2018) a flexible new load, power-to-x, eats the surplus storage would have stored, and the planner builds a small fraction of the storage. An option valued on its own overstates what it adds to a system with the others.

Closing line: what it all adds up to.

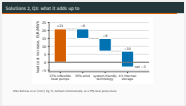
Solutions 2, Q3: the questions

Q3. Fig. 13 and “overcompensate”. Do heat pumps help integrate wind, or does wind help integrate heat pumps?

Solutions 2, Q3: what it adds up to



After Ruhnau et al. (2020), Fig. 13, redrawn schematically: at a 15% heat pump share.



Solutions 2, Q3: what it adds up to

Lead line: wind, system-friendly technology and storage together more than offset the load cost increase: it is a property of inflexible heat pumps in a system without wind, not of electrified heat.

- **“Overcompensate” (p. 11):** the load cost ends a little below that of the first, inflexible heat pumps.
- **Each remedy is a choice:** of wind, of technology, of a store.

Closing line: so, do heat pumps help integrate wind?

Solutions 2, Q3: heating with wind

- **Q3. Heat pumps hardly help wind:** they raise its value only at a fixed wind capacity, and neither better technology nor storage changes that.
- **Q3. Wind helps heat pumps:** it lowers their load cost, and with the two remedies more than offsets its increase.
- **Q3. Only if someone chooses the remedies:** on fixed tariffs and standard load profiles nobody is paid to (p. 12); hence settlement on wholesale prices through smart meters.


Solutions 2, Q3: heating with wind

Q3. Heat pumps hardly help wind: they raise its value only at a fixed wind capacity, and they change the optimal capacity too—by more than the...

Q4. Wind helps heat pumps: it makes them cheaper to run, and with the two remedies, more: it flattens its revenue.

Q5. Only if ruminism chooses the remedies, on fixed tariffs and standard load profiles nobody is paid (p. 12), hence settlement: an wholesale prices through smart meters.

Solutions 2, Q3: heating with wind

Lead line: the help runs from wind to heat pumps, not the other way.

- **Standard load profiles:** every kWh costs the same whenever it is used, so “the cost and benefit of volatility and flexibility are not internalized but socialized” (p. 12). The owner is paid for fewer kWh, not for a steadier profile or a store.
- **The four ways to balance** compete for the same two tails: support for one may crowd out another rather than add to the total.

Closing line: three things to carry forward.

Wrap-up

What carries forward

1. **Trade relieves both tails when the weather differs across the wire.** A cable is paid by the hours it is full; its value is concave, and its gains are unevenly shared.
2. **A new load has a load cost,** the mirror of a capture price. Heat pumps that follow the heat load consume in the scarcest hours, and cost more the more there are; wind, steadier heat pumps and storage bring it back down.
3. **Flexibility competes with flexibility.** Heat pumps do little for wind once wind grows with them, and thermal storage crowds out cables and pumped hydro.

Lecture 6 lets the planner build the whole fleet: investment.

Before lecture 6 (Monday 12 October): *Energy System Models*, section 7 and appendix B; Brown et al. (2025). Full solutions to today's worksheet go up after class.

References i

Brown, Tom, Fabian Neumann, and Iegor Riepin (2025). **“Price Formation Without Fuel Costs: The Interaction of Demand Elasticity with Storage Bidding”.** *Energy Economics* 146, p. 108483.

Gonzales, Luis E., Koichiro Ito, and Mar Reguant (2023). **“The Investment Effects of Market Integration: Evidence from Renewable Energy Expansion in Chile”.** *Econometrica* 91.5, pp. 1659–1693.

Hogan, William W. (2018). **“A Primer on Transmission Benefits and Cost Allocation”.** *Economics of Energy & Environmental Policy* 7.1, pp. 25–45.

Joskow, Paul L. and Jean Tirole (2005). **“Merchant Transmission Investment”.** *The Journal of Industrial Economics* 53.2, pp. 233–264.

Ruhnau, Oliver, Lion Hirth, and Aaron Praktiknjo (2020). **“Heating with Wind: Economics of Heat Pumps and Variable Renewables”.** *Energy Economics* 92, p. 104967.

Zerrahn, Alexander, Wolf-Peter Schill, and Claudia Kemfert (2018). **“On the Economics of Electrical Storage for Variable Renewable Energy Sources”.** *European Economic Review* 108, pp. 259–279.